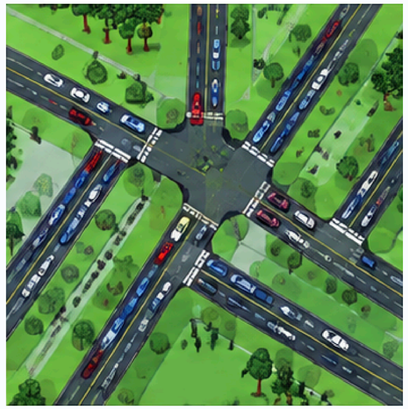
Traffic Simulation Environment (SUMO)

SUMO



Traffic Observation

Reward Computation

Dynamic Vehicle Environment

Vehicle Clients (FLDQN Agents)

Vehicle 1

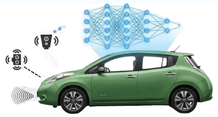


FLDQN
Local Training
(DQN Policy)

Privacy Mechanism
(CPQ / CPQ-λ / LDP-FL
Gaussian / Laplace)

PQC Signing
(ML-DSA / SLH-DSA / Falcon)

Vehicle 2



FLDQN
Local Training
(DQN Policy)

Privacy Mechanism
(CPQ / CPQ-λ / LDP-FL
Gaussian / Laplace)

PQC Signing
(ML-DSA / SLH-DSA / Falcon)

Vehicle N



FLDQN
Local Training
(DQN Policy)

Privacy Mechanism
(CPQ / CPQ-λ / LDP-FL
Gaussian / Laplace)

PQC Signing
(ML-DSA / SLH-DSA / Falcon)

Communication

ns-3
NETWORK SIMULATOR



RSU / Aggregation Server

PQC Signature Verification

- Verify sender
- Ensure integrity
- Reject invalid updates

Parameter Reconstruction

- Unpack parameters
- Restore tensor shapes

Federated Aggregation

- Aggregate verified updates
- Produce global model